Kernel: SageMath 10.7

Academic Year 2025-26

Branch: Industrial IOT

Section : K

Batch : K-1

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Aim:To evaluate definite integrals by employing Sage Math using Numerical Techniques

#Procedure (Step-by-Step Commands in Sage):

1. To evaluate a given definite integral using the Trapezoidal Rule.

2. To evaluate a given definite integral using Simpson's 1/3 Rule.

3. To evaluate a given definite integral using Simpson's 3/8 Rule using Sage Math.

```
#Step 1: Declare the variable
var('x')
```

Out[1]: x

```
#Step 2: Define the given function
f(x) = x^3 - 4*x + 1
show("Given function f(x) =", f)
```

Out[2]: Given function f(x) $= x \mapsto x^3 - 4x + 1$

```
#Step 3: Define the limits of integration
a = 0
b = 2
```

```
#Step 4: Apply Trapezoidal Rule
n = 10
h = (b - a)/n
T = (h/2)*(f(a) + f(b) + 2*sum(f(a + i*h) for i in range(1, n)))
show("Value of integral using Trapezoidal Rule =", T)
```

Out[4]: Value of integral using Trapezoidal Rule $= -\frac{49}{25}$

```
#Step 5: Apply Simpson's 1/3 Rule
S13 = (h/3)*(f(a) + f(b) + 4*sum(f(a + (2*i - 1)*h) for i in range(1, n//2 + 1))
+ 2*sum(f(a + 2*i*h) for i in range(1, n//2)))
show("Value of integral using Simpson's 1/3 Rule =", S13)
```

Out[5]: Value of integral using Simpson's 1/3 Rule $= -2$

```
#Step 6: Apply Simpson's 3/8 Rule
n = 6
h = (b - a)/n
S38 = (3*h/8)*(f(a) + f(b)+ 3*sum(f(a + i*h) for i in range(1, n) if i % 3 !=
0)+ 2*sum(f(a + i*h) for i in range(3, n, 3)))
show("Value of integral using Simpson's 3/8 Rule =", S38)
```

Out[6]: Value of integral using Simpson's 3/8 Rule $= -2$

Example for practice

```
#1. Evaluate ∫ (x^2 + 2x) dx using Simpson's 1/3 Rule a=3 b =1
var('x')
f(x) = x^2 + 2*x
show("Given function f(x) =", f)
a = 1
b = 3
n = 8
h = (b - a)/n
S13 = (h/3)*(f(a) + f(b) + 4*sum(f(a + (2*i - 1)*h) for i in range(1, n//2 + 1))
+ 2*sum(f(a + 2*i*h) for i in range(1, n//2)))
show("Value of integral using Simpson's 1/3 Rule =", S13)
```

Out[7]: Given function f(x) =$x \mapsto x^2 + 2x$

Value of integral using Simpson's 1/3 Rule =$\frac{50}{3}$

```
#example2
var('x')
f(x) = 2.71828^x
show("Given function f(x) =", f)
a = 0
b = 1
n = 7
h = (b - a)/n
T = (h/2)*(f(a) + f(b) + 2*sum(f(a + i*h) for i in range(1, n)))
show("Value of integral using Trapezoidal Rule(using e value)=", T)
```

Out[8]: Given function f(x) =$x \mapsto 2.71828000000000^x$

Value of integral using Trapezoidal Rule(using e value)=1.72120240526268

```
#example 3
var('x')
f(x) = x^3 + 1
show("Given function f(x) =", f)
a = 0
b = 2
n = 9
h = (b - a)/n
S38 = (3*h/8)*(f(a) + f(b)+ 3*sum(f(a + i*h) for i in range(1, n) if i % 3 !=
0)+ 2*sum(f(a + i*h) for i in range(3, n, 3)))
show("Value of integral using Simpson's 3/8 Rule = ", S38)
```

Out[9]: Given function f(x) =$x \mapsto x^3 + 1$

Value of integral using Simpson's 3/8 Rule =6

Result : The definite integral of given function is successfully evaluated using Trapazoidal Rule,Simpson's 1/3rd Rule, and Simpson's 3/8th Rule with the help of SageMath.